

Harshil Gorasiya

LLM systems: retrieval, agents, and small-model research

LLM Applications · RAG · AI Agents · LLM Research

Heilbronn, Germany | +49 155 6351 7346 | harshil.gorasiya.0011@gmail.com
LinkedIn: linkedin.com/in/harshil-gorasiya-054a58319
GitHub: github.com/N0t-Harshil | Portfolio: harshil-0011.github.io/My-Portfolio

Seeking: Working student position (20 hrs/week during semester) · Full-time during semester breaks · LLM/RAG role after graduation (2027)
Available: immediately | **Residence status:** valid residence permit with work permit

0.654	8.1 ms	28–32 FPS	79 %	250 tests
nDCG@10, BEIR SciFact, 300 queries	overhead per agent step, laptop CPU	YOLOv8n on CPU only	accuracy (up from 71 %)	pytest suite, Ekdant

PROFILE

Master's student at Heilbronn University focused on LLM applications. I build local RAG systems with hybrid retrieval and AI agents with tool use, and I research a reasoning module for language models in a supervised project. I also bring two ML internships and a role in AI-enabled web development.

EDUCATION

since 03/2025 — **M.Sc. Software Engineering and Management** | Hochschule Heilbronn
English-taught programme | GPA so far: 1.9 (German scale) | expected graduation: 2027

2020 – 2024 — **B.Tech. Information Technology** | A. D. Patel Institute of Technology, Gujarat
India | Grade: 1.9 (converted to German scale)

PROJECTS

AI agent — **Ardan-CLI: Autonomous Terminal Coding Agent**
TypeScript (strict), Node.js 24, 0 runtime dependencies, MCP, LSP

- Agent loop with streaming, **35 tools**, plan mode, sub-agents, and checkpoints; **8 LLM providers** with automatic failover.
- SSE/NDJSON streaming, MCP and LSP clients without SDKs; fallback parser for models without function calling.
- **8.1 ms** overhead per agent step (laptop CPU); CI on Linux, macOS, Windows.

Research engine — **Local Perplex: Local AI Research Assistant with Citations**
Python, FastAPI, SSE, SQLite, Ollama, Docker

- Searches the web and your own documents (PDF, Word, Excel, and more), reads pages in parallel, and streams cited answers; **26 LLM providers**, 7 search backends, fully local if desired.
- Own BM25 ranker: **0.654 nDCG@10** on BEIR SciFact, 300 queries (previous version 0.038; Anserini reference 0.665), 9.5 ms per query.
- Answer page in about **200 ms**; SQLite knowledge graph (77× faster writes); **55 tests**, 76 % coverage, CI on Ubuntu and Windows.

RAG system — **ANVITHA: Local Knowledge Vault with Grounded, Cited Answers**
Python, FastAPI, Ollama, SQLite, NumPy, BM25

- Turns PDF, TXT, and Markdown files into a linked Markdown knowledge vault; answers come only from the vault and cite the original page.
- Facts are checked against the source text before they are stored; every answer sentence is checked against the vault afterwards, and unsupported parts are removed.
- Questions outside the vault are refused in under 0.3 s without a model call.
- Tested with a local 0.8B-parameter model (Ollama): 8 of 8 covered questions answered (12 test questions in total), median 7.4 s to full answer; 26 automated tests in about 1.5 s.

Master's thesis (ongoing)

Ekdant: Debate-Sandbox Reasoning Module for Language Models
Hochschule Heilbronn, supervised by a professor | Python, PyTorch, Hugging Face

- Research question: does a parallel debate module (4 trajectories, 2 debate rounds, consensus) improve a language model's reasoning? Implemented in pure PyTorch as a zero-initialized residual module (**+19.59M parameters**) on a pretrained SmolLM2-135M.
- Controlled A/B comparison against an architecture-matched baseline on generative medical QA (PubMedQA, MedQuAD); leakage-free split by question; evaluated with perplexity, ROUGE-L, token F1, and paired-bootstrap confidence intervals.
- Best variant (module as final stage) beats the baseline on every metric, but only narrowly (perplexity **3.783 to 3.776**, one run); the from-scratch pilot showed no difference. Second seed planned to confirm.
- **250 pytest tests**; found and fixed 4 design defects; literature base of 66 papers.

More projects

AI Video Upscaling | Python, PyTorch
Windows desktop application for video super-resolution with model selection and runtime monitoring; compared Real-ESRGAN variants using PSNR and SSIM.

MNIST-MLOps | Python, PyTorch, Flask, SQLite
Demo of an MLOps workflow: incremental training, metrics and model versions in SQLite, web dashboard.

PROFESSIONAL EXPERIENCE

05/2024 – 02/2025 (10 months)

AI & Web Solutions Developer | Aani Brothers Infotech, Surat, India

- Built web applications that translate business requirements into user interfaces and application workflows.
- Contributed to web projects with AI-enabled features; integrated functional requirements into the user experience.
- Conducted user research, wireframing, and prototyping; iterated on designs to improve usability and consistency.

12/2023 – 04/2024 (5 months)

Deep Learning Intern | PD Avenue, Surat, India

- Built a residual-block CNN (TensorFlow/Keras) for age and gender classification on **22,000+ images**; raised validation accuracy from **71 % to 79 %** (architecture optimization, class-balanced augmentation).
- Cut training time on V100 GPUs by **20 %** (cosine learning-rate scheduling, larger batch size) with no loss of accuracy.
- Found and fixed a data-leakage issue; ensured reliability across **10+ structured training runs**.

05/2023 – 06/2023 (2 months)

Machine Learning Intern | Dotcom IoT LLP, Surat, India

- YOLOv8n object detection on a CPU-only edge device (Intel Core i7): 720p video at **28–32 FPS**; **2.4× higher throughput** than the YOLOv5s baseline.
- Curated and annotated **3,000+ images** across five vehicle classes.
- Tuned IoU and NMS parameters: reduced false positives in low light by **18 %** (dedicated night-test dataset).
- Packaged inference and visualization into a deployment-ready Python application with performance benchmarking.

SKILLS

LLM & RAG	RAG pipelines, hybrid retrieval (BM25, embeddings), grounded answering, Ollama, ReAct agents and tool use, Anthropic API, Model Context Protocol (MCP), prompt engineering
LLM research	Fine-tuning, Transformer architectures (GQA, SwiGLU), A/B experiment design, PyTorch, Hugging Face Transformers
ML & Vision	PyTorch, TensorFlow/Keras, scikit-learn, YOLOv8, CNNs, OpenCV
Engineering	Python, TypeScript, SQL, Bash, Git, Docker, Linux, pytest, CI (GitHub Actions), REST APIs, FastAPI, Flask, MLflow, SQLite, PostgreSQL, MongoDB, AWS (basic)
Evaluation	Latency and throughput benchmarking, nDCG@10 (BEIR), validation accuracy, false-positive analysis, PSNR/SSIM, perplexity, ROUGE-L, token F1, paired bootstrap

CERTIFICATIONS

2026	Claude with the Anthropic API | Model Context Protocol: Advanced Topics (both Anthropic)
2025	Machine Learning with Python (IBM) | Fundamentals of Data Science (Databricks)
Further	CRISP-DM on the AWS Stack | Big Data with Spark and Hadoop | AWS Academy Cloud

LANGUAGES

English: fluent | **German:** A2–B1 (self-assessed) | **Hindi:** fluent | **Gujarati:** native

Heilbronn, 05.11.2026